

WorldCore: Consistent Video World Model with Implicit Memory Representation

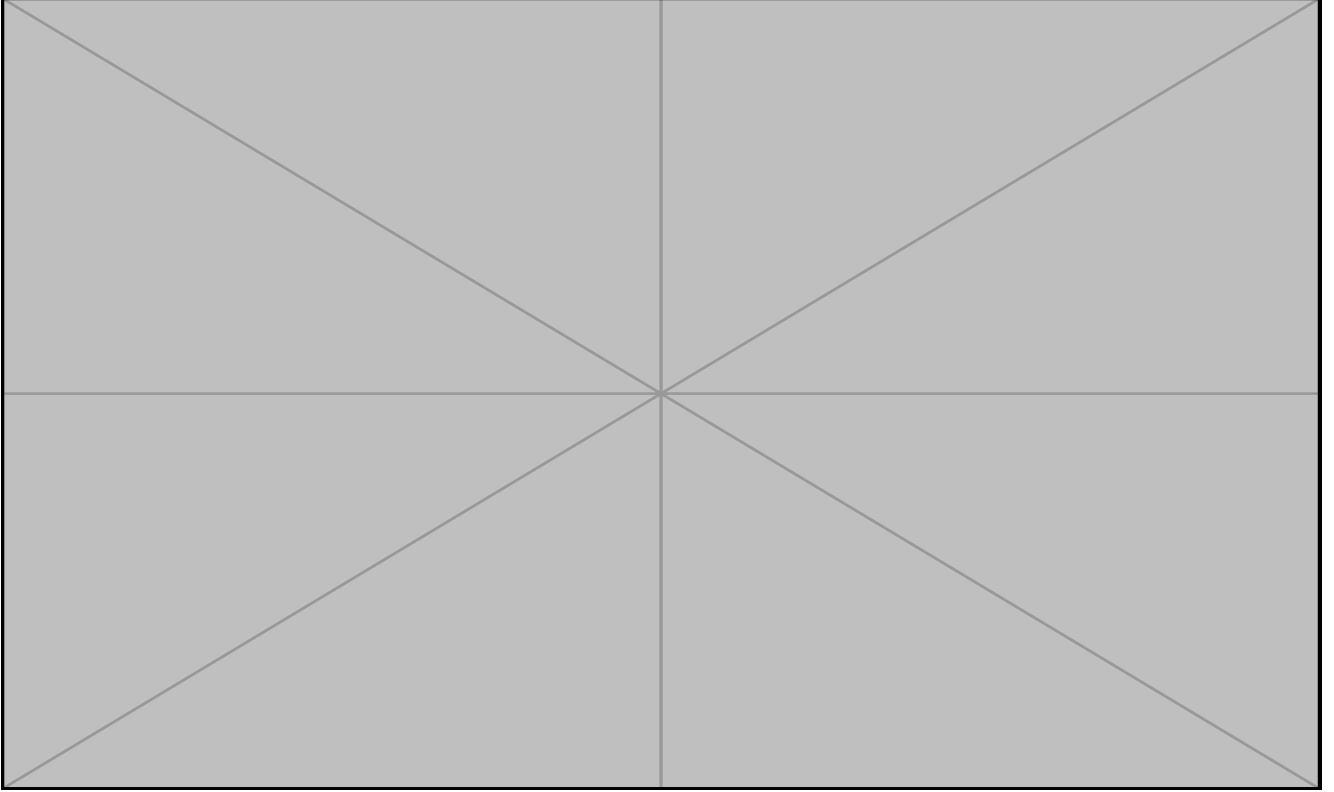


Figure 1. Teaser placeholder.

Abstract

Video world models transform generative video into interactive simulations, yet sustaining a coherent world over long-horizon rollouts remains challenging once earlier observations leave the finite context window. Existing memory mechanisms either retain selected history with an inherent trade-off between coverage and efficiency or rely on 3D representations that are unreliable in dynamic scenes or difficult to reconcile with the generative space of video diffusion models. Our key insight is that persistent world modeling calls for a compact 3D-aware memory learned jointly with the video generator. We therefore present **WorldCore**, a video world model that encodes history latent frames into a compact 3D-aware implicit memory representation. The

memory encoder is initialized from pretrained multi-view encoders to inherit a learned 3D inductive bias and jointly optimized with the video generator, allowing the memory space to be learned alongside the DiT token space. A fixed-budget memory update mechanism dynamically updates the representation based on user input, while camera-conditioned autoregressive generation and few-step distillation yield streaming inference at 20 FPS on two H200 GPUs. Experiments demonstrate improved long-range consistency and generalization to unseen environments while preserving precise camera control and visual quality, including minute-scale rollouts under complex user-specified camera trajectories.

1. Introduction

Video world models aim to simulate the world through generative video modeling [13, 22, 31, 43, 55, 57]. Unlike conventional video generators that produce a fixed visual sequence, a video world model causally predicts how observations evolve under a stream of events. This formulation turns video generation into an interactive simulation process in which every predicted frame becomes part of the environment that conditions what happens next. By exposing a simulated world through video, a visual medium that closely reflects how humans perceive and reason about their surroundings, such models provide a scalable interface for embodied intelligence, autonomous driving, physical simulation, content creation, and game industry [14, 19, 36, 47, 65, 68].

Building on rapid advances in video generation, video world models have quickly expanded from simple games to open domain, with recent systems advancing flexible interaction, real-time generation, and long-duration rollout [4, 10, 15, 32, 41, 45, 54, 63, 87]. Across these developments, a fundamental requirement is a persistent state representation that gives the model memory and anchors successive predictions to a consistent world. This state representation must retain what has been observed after it leaves, then recover the previous information in later revisit.

A straightforward way to preserve memory for video world models is to involve their previously generated frames in attention computing. Keeping the full history frames gives the model access to every past observation, but incurs substantial storage and attention costs [5, 18, 72]. Some solutions retrieve only a few relevant history frames [5, 7, 34, 51, 52, 63, 66, 69, 72, 75, 77] to reduce computational overhead and support real-time inference, which inevitably face a trade-off between coverage and efficiency and are prone to inconsistency under complex camera trajectories. Rather than storing history frames, explicit spatial memories like point clouds provide a shared 3D reference across viewpoints. Nevertheless, they rely on accurate geometry estimation and become unreliable when modeling moving objects and dynamic environment [67, 74, 80, 86]. Another line of work encodes history frames into implicit memory representations [50, 81], with recent methods further incorporating features from geometry estimation models [58] to improve 3D-awareness for these representations [21, 64]. However, aligning feature spaces learned for geometry estimation with the token space of video diffusion models remains challenging. Consequently, existing methods [21, 64] remain limited to a small number of training scenes and generalize poorly to unseen environments.

Recent advances in 3D representation learning [26–28, 53] have demonstrated remarkable capabilities in learning 3D-aware scene representations. These representations are inherently compact while retaining rich geometry and

appearance information, making them a natural starting point for the state representation of video world models. Motivated by this, we present **WorldCore**, a video world model with implicit memory representation. At its core, a memory encoder initialized from pretrained encoders for 3D representation learning maps history latent frames into a compact 3D-aware memory space, transferring a learned 3D inductive bias to the resulting space. We optimize this memory space by jointly training the memory encoder and video generator, co-adapting it with the video DiT token space, achieving consistent and long-horizon world modeling.

To reduce computational overhead and support real-time interaction, we maintain a fixed memory budget and introduce a memory updating mechanism that dynamically write and read the memory representation based on user input. Together with camera-conditioned autoregressive generation and few-step distillation, our model achieves streaming inference at 20 FPS on two H200 GPUs while preserving precise camera control and minute-scale consistency under complex user-specified camera trajectories.

WorldCore makes the following contributions:

- We explore learning a 3D-aware implicit memory representation for video world models to improve consistency over long-horizon rollouts.
- We integrate the 3D-aware memory representation into camera-controllable autoregressive video generation, achieving long-range consistency and precise camera control under complex user-specified trajectories within a fixed memory budget.
- We further build a real-time interactive system that streams at 20 FPS on two H200 GPUs while preserving long-range consistency and visual quality.

2. Related Work

2.1. Interactive Video World Models

Video world models generate future observations in response to user actions, turning video generation into interactive simulation [22]. Recent methods increasingly combine streaming generation with camera control to support real-time interaction and long-horizon exploration [1, 4, 9, 11, 12, 15, 25, 41, 42, 49, 51, 52, 63, 74, 85, 87].

Streaming generation methods broadly follow two paradigms. The first restructures the denoising process for causal rollout using moving FIFO queues or temporally varying noise levels [6, 29], whereas the second adapts pretrained bidirectional teachers through causal distillation and self-forcing training [8, 23, 40, 71, 73]. Despite their differences, both paradigms autoregressively generate fixed-length chunks and reuse recent outputs as context [17]. Such recent context cannot recover content that has remained outside the context window. Consequently, existing

streaming generation methods lack a persistent world state for maintaining long-range consistency.

Camera control is commonly implemented through discrete action inputs [10, 32, 63, 76], continuous camera parameters [3, 16, 35, 61, 82], or point-cloud renders along the target trajectory [48, 78, 79]. Although these signals specify viewpoint changes, they do not provide a persistent state that preserves scene content beyond the context window. Thus, combining streaming generation with camera control alone remains insufficient for consistent long-horizon exploration.

2.2. Memory Mechanisms in Video World Models

Persistent video world models require memory beyond the recent video context. We categorize existing approaches according to their stored representations: context memory, spatial memory, and implicit memory.

Context memory directly retains history frames into attention computation [5, 18, 44, 52, 63, 69, 75, 77, 83]. To operate within a limited context budget, these methods retrieve only a few history frames as memory. Geometry-aware variants further select past history frames based on camera overlap or reconstructed surfaces [20, 34, 51]. Their performance is therefore constrained by geometry quality or context capacity, resulting in an inherent trade-off between coverage and efficiency.

Spatial memory instead transforms history frames into views specified by target camera poses [33, 48, 62, 67, 70, 74, 79, 80, 86]. Representative methods perform this transformation via novel view synthesis, encode the synthesized target-view frames into the VAE latent space, and concatenate the resulting latents channel-wise with the input noise to condition video diffusion models [33, 48, 62, 67, 79]. The resulting spatial correspondence across viewpoints facilitates revisiting previously observed regions. However, strong alignment to the target view can overconstrain scene dynamics, limiting their ability to model dynamic objects and environments.

Implicit memory encodes history frames into a compact learned state [7, 37, 50, 66, 81]. Existing methods either update this state recurrently alongside a local context window [50, 81] or compress observations using a learned encoder [7, 66]. Recent approaches [21, 64] incorporate features from frozen VGGT [58] to improve 3D awareness, either as direct context for video diffusion models or as alignment targets for encoder outputs. However, the substantial gap between VGGT geometry features and the DiT token space makes alignment difficult, causing these methods to overfit a few training scenes and generalize poorly. Our method follows the encoder-based implicit memory paradigm and initializes the memory encoder from pre-trained multi-view encoders for 3D representation learning [26, 28, 53], inheriting a learned 3D inductive bias. By

jointly optimizing the memory encoder and video generator, we learn the 3D-aware memory space alongside the DiT token space, supporting generalizable and consistent long-horizon world modeling under complex camera trajectories.

3. Method

3.1. Preliminary

Latent video diffusion. Let $\mathbf{x} \in \mathbb{R}^{3 \times F \times H \times W}$ denote a clean video of F frames. A video variational autoencoder (VAE) [30] maps it to a spatiotemporal latent $\mathbf{z} = \mathcal{E}_{\text{VAE}}(\mathbf{x}) \in \mathbb{R}^{c \times f \times h \times w}$. Its decoder reconstructs the video as $\hat{\mathbf{x}} = \mathcal{D}_{\text{VAE}}(\mathbf{z})$. Operating in this latent space substantially reduces the token sequence processed by the Diffusion Transformer (DiT) [46]-based denoiser, which patchifies $\mathbf{z}$ into video tokens and applies 3D self-attention. In a conventional bidirectional video diffusion model such as Wan 2.1 [57], every video token can attend to all other tokens in the clip during each denoising step.

The denoiser is trained with conditional flow matching [39]. For diffusion time $t \sim \mathcal{U}(0, 1)$ and Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, the noisy latent and its target velocity are

$$\mathbf{z}_t = (1 - t)\mathbf{z} + t\epsilon, \quad \mathbf{v}_t^* = \epsilon - \mathbf{z}. \quad (1)$$

Given a text condition $\mathbf{y}$, the DiT $\mathbf{v}_\theta$ predicts this velocity by minimizing

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{z}, t, \epsilon} \left[\|\mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{y}) - \mathbf{v}_t^*\|_2^2 \right]. \quad (2)$$

Chunk-wise autoregressive video generation. To extend generation beyond a fixed clip, autoregressive methods divide a long video into fixed-length chunks and generate them sequentially. At a generic rollout step, we omit the rollout index from all quantities for clarity. Let $\mathbf{z}$ denote the clean latent of the current chunk, $\mathbf{z}_t$ its state at diffusion time t , $\mathbf{z}^h$ the accumulated clean history frames, and $\mathbf{z}^r$ the fixed-length recent history frames retained from $\mathbf{z}^h$. Given the text condition $\mathbf{y}$, a standard chunk-wise autoregressive model evolves the current latent through the conditional flow

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{z}^r, \mathbf{y}). \quad (3)$$

At each denoising step, the DiT processes the concatenated latent sequence $[\mathbf{z}^r; \mathbf{z}_t]$, where $[\cdot; \cdot]$ denotes concatenation along the token sequence. After denoising, $\mathbf{z}$ is appended to $\mathbf{z}^h$, and the sliding window of $\mathbf{z}^r$ is updated with $\mathbf{z}$.

3.2. Base Architecture

To enable long-horizon memory and user interaction, we extend the standard autoregressive flow in Eq. (3) by conditioning each chunk jointly on a memory $\mathbf{M}$ derived from

the accumulated history $\mathbf{z}^h$ and a target camera trajectory $\mathbf{C}$:

$$\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_\theta(\mathbf{z}_t, t \mid \mathbf{M}, \mathbf{z}^r, \mathbf{C}, \mathbf{y}). \quad (4)$$

Memory conditioning. At each denoising step, the DiT processes $[\mathbf{M}; \mathbf{z}^r; \mathbf{z}_t]$ as a single latent sequence. This adds a dedicated memory stream to the recent history while preserving chunk-level causality. Unlike prior context-based memory [52, 63, 69, 75] that instantiate $\mathbf{M}$ as a fixed-length context retrieved from the accumulated history $\mathbf{z}^h$, we model $\mathbf{M}$ by mapping $\mathbf{z}^h$ into a compact 3D-aware implicit memory representation.

Camera conditioning. The target trajectory $\mathbf{C}$ specifies a camera-to-world pose and camera intrinsics for each frame. Following PRoPE [35], we encode relative camera geometry as a positional transformation within self-attention. We implement this conditioning using the parallel camera-attention branch of UCPE [82], which adopts independent query, key, and value projections and adds its output to the original self-attention through a zero-initialized projection. The camera branch is applied only to the noisy part $\mathbf{z}_t$ of the concatenated sequence; the memory $\mathbf{M}$ and recent context $\mathbf{z}^r$ are processed without camera injection.

3.3. Memory Encoder

Memory writing. After generating the first chunk, the memory encoder Φ writes the accumulated history latents $\mathbf{z}^h$ and their corresponding camera parameters $\mathbf{C}^h$ into an implicit 3D-aware representation $\mathbf{R}$:

$$\mathbf{R} = \Phi(\mathbf{z}^h, \mathbf{C}^h) \in \mathbb{R}^{|\mathbf{z}^h|L \times d}, \quad (5)$$

where $|\mathbf{z}^h|$ denotes the number of history latent frames. The encoder produces L tokens of dimension d per history latent frame.

We initialize the encoder architecture and weights from the LagerNVS encoder [53], discarding its shallow image-processing layers and adding a new patch embedding layer to map each latent frame directly into its representation space. The history camera poses are expressed relative to the latest latent frame in $\mathbf{z}^h$ and injected as camera tokens during encoding. The resulting $\mathbf{R}$ aggregates geometry and appearance information across the input history without materializing an explicit 3D reconstruction.

Since the length of $\mathbf{R}$ grows linearly with the number of input history frames $|\mathbf{z}^h|$, to bound the cost of the encoder, we restrict its input to k history latent frames. At inference, we retain the latest latent frame in $\mathbf{z}^h$ and greedily select $k - 1$ complementary frames whose joint FoV maximizes coverage of the target region along the upcoming camera trajectory. We denote the selected history latents and their camera parameters by $\mathbf{z}^s$ and $\mathbf{C}^s$, respectively. Prior context-based memory methods [52, 63, 69, 75] retrieve only a few history frames by ranking their pairwise FoV

similarity to target poses [63, 75], resulting in limited coverage and sensitivity to individual selections. In contrast, our memory encoder accommodates more history frames under a comparable budget, while max-coverage history retrieval yields broader coverage and greater robustness.

Memory readout. The written representation $\mathbf{R}$ should then be read out as a fixed-size memory $\mathbf{M}$ that conditions the DiT. We compare two readout mechanisms under a fixed DiT token budget.

The first is pose-free readout, in which a learned readout module maps the complete representation into a fixed set of memory tokens compatible with the DiT input:

$$\mathbf{M} = \text{Readout}(\Phi(\mathbf{z}^s, \mathbf{C}^s)). \quad (6)$$

This readout manner is independent of the upcoming camera trajectory, leaving the DiT attention to identify information relevant to the current generation.

The second is pose-guided readout, which queries the representation using a fixed-size subset $\mathbf{C}^q \subset \mathbf{C}$ sampled from the upcoming target camera trajectory:

$$\mathbf{M} = \text{Readout}(\Phi(\mathbf{z}^s, \mathbf{C}^s), \mathbf{C}^q). \quad (7)$$

In pose-guided readout, we initialize the readout module with the shallow decoder layers of LagerNVS [53] and add projection layers to map the output tokens into the DiT token space.

Compared with the pose-free readout, the pose-guided readout allocates the fixed memory budget to information relevant to the upcoming target poses. Empirically, We it converges more reliably. We therefore adopt pose-guided readout, with ablations in Sec. 4.

3.4. Dataset Curation and Base Model Training

3.5. Distillation for Real-time Interaction

4. Experiments

4.1. Experimental Setup

Benchmark. We construct a unified benchmark for long-horizon, camera-controllable video world models, evaluating revisit memory, camera control, and visual quality separately. It contains 145 initial scenes sampled from HappyOyster [15], Project Genie, and Pexels. Of these, 83 are agent-centric; the remaining 62 span other visual domains. Each initial image and text description is paired with five metric camera trajectories, yielding $145 \times 5 = 725$ videos per method. The first four trajectories last 33.0–49.5 seconds and cover standard and composite camera motion. Trajectory 05 contains 1,650 target poses over 103.1 seconds and retraces its first half in reverse, providing exact closed-loop revisits for memory evaluation. Table 1 summarizes their construction and use.

Table 1. Benchmark trajectories. The first four cover standard and composite camera motion; Trajectory 05 is a long closed-loop revisit.

Trajectory	Motion pattern	Poses	Duration	Camera	Memory	VBench
01	backward, turn, forward, and lateral motion	594	37.1 s	✓		✓
02	four right-turn and forward-motion groups	792	49.5 s	✓		✓
03	repeated forward and right-turn motion	528	33.0 s	✓		✓
04	left turn, long forward motion, then right turn	792	49.5 s	✓		✓
05	50-chunk forward-reverse palindrome	1,650	103.1 s	✓	✓	✓

Table 2. Baseline configurations and the common evaluation format. Each method generates five trajectories for 145 scenes, giving 725 videos.

Method	Configuration	Native resolution	Evaluation resolution	Frame rate
WorldCore	multi-step, without DMD	640×384	640×384	16 fps
DreamX-World [11]	metric-pose configuration	1280×704	640×384	16 fps
EVOKE [74]	DMD configuration	640×384	640×384	16 fps
HY-WorldPlay [52]	distilled configuration	832×480	640×384	16 fps
JoyAI-Echo [85]	Echo-WM-Flash	640×384	640×384	16 fps
LingBot-World-v2 [12]	fast configuration	800×480	640×384	16 fps
Matrix-Game 3.5 [49]	distilled FP	1280×704	640×384	16 fps
SANA-WM [87]	standard world-model configuration	1280×704	640×384	16 fps

Baselines and video normalization. We compare WorldCore with seven recent camera-controllable world models: DreamX-World [11], EVOKE [74], HY-WorldPlay [52], JoyAI-Echo [85], LingBot-World-v2 [12], Matrix-Game 3.5 [49], and SANA-WM [87]. Methods that accept continuous camera poses receive the same metric pose sequence; for other control interfaces, we convert this sequence to the native action representation. All methods use identical initial images, descriptions, and target trajectories. We neither resample random seeds nor filter generated outputs. As shown in Table 2, we evaluate WorldCore without DMD distillation and use the standard accelerated configuration available for each baseline. Before evaluation, every output is deterministically resized to 640×384 and resampled by timestamp to 16 fps, without cropping or padding. All metrics read the same normalized RGB video.

Evaluation protocol. Revisit memory is evaluated only on the closed-loop Trajectory 05, camera control uses all five trajectories, and VBench uses every complete video. For memory and camera metrics, frame 0 is aligned with target pose 0. For VBench, we remove the repeated conditioning frame. WorldCore produces 593, 789, 525, 789, and 1,649 frames on the five trajectories, with missing frames confined to the sequence ends; memory pairs and stride-4 camera samples are restricted to indices valid for every method. VBench processes each complete valid output, totaling 629,300 frames for WorldCore and 630,895 for each baseline. Metrics are first aggregated within a scene and then averaged equally across 145 scenes. This pre-

vents long trajectories from dominating the reported score through densely sampled neighboring frames.

4.2. Long-Horizon Revisit Memory

Protocol. Trajectory 05 forms a half-open palindrome of length $N = 1650$. For an outbound frame $i \geq 1$, the frame with the same target pose on the return path is

$$j = N - i. \quad (8)$$

We sample $i \in \{2, 6, 10, \dots, 822\}$ and search within ± 3 frames of its nominal match j for the pose with the smallest translation and rotation error. All retained matches pass the pose-consistency check. This dense stride-4 protocol gives 206 pairs per scene and 29,870 pairs overall, covering the complete outbound and return paths rather than only their endpoints.

Metrics. We report MET3R [2], LPIPS [84], PSNR, and SSIM [60]. MET3R measures 3D-aware feature consistency after dense alignment; we resize frames to 256×256 and use the MAST3R backbone, DINO-16 features, FeatUp upsampling, and cosine distance. LPIPS uses an AlexNet backbone on the full 640×384 frames. PSNR is computed in 8-bit RGB, while SSIM uses an 11×11 uniform window with the standard constants $C_1 = (0.01 \times 255)^2$ and $C_2 = (0.03 \times 255)^2$. Lower MET3R and LPIPS indicate better consistency; higher PSNR and SSIM are better.

Results. WorldCore ranks first on all four revisit metrics in Table 3. It reduces MET3R from the second-best 0.3953 to 0.2094 and LPIPS from 0.5124 to 0.3090. It also raises

Table 3. Long-horizon revisit consistency on Trajectory 05. Results average 29,870 matched frame pairs. Dark and light blue mark the best and second-best results.

Method	MEt3R↓	LPIPS↓	PSNR↑	SSIM↑
WorldCore	0.2094	0.3090	16.2242	0.4451
DreamX-World	0.5627	0.6307	12.4322	0.2143
EVOKE	0.4267	0.5662	11.9073	0.2615
HY-WorldPlay	0.3953	0.5124	12.6642	0.2298
JoyAI-Echo	0.4706	0.5873	12.1153	0.2025
LingBot-World-v2	0.5080	0.6349	10.1678	0.2079
Matrix-Game 3.5	0.4113	0.5505	12.7984	0.2135
SANA-WM	0.4033	0.5544	12.7554	0.2234

PSNR from 12.7984 to 16.2242 dB and SSIM from 0.2615 to 0.4451. The consistent improvement across all four metrics indicates that WorldCore retains previously observed content over the 103-second closed loop.

4.3. Camera Control

Relative trajectory evaluation. We recover camera trajectories from generated videos using the VGGT-Ω-1B-512 model [59], which supports static and dynamic scenes. Frames are sampled every four steps, corresponding to 0.25 seconds at 16 fps, and each complete video is processed in one global inference pass. Predicted and target poses are converted to camera-to-world coordinates and normalized by their first sampled pose, $P_t^{\text{rel}} = P_0^{-1} P_t$. We align predicted camera centers to their targets with Umeyama Sim(3) [56] and apply one global rotation alignment. For target pose $P_t = [R_t \mid \mathbf{t}_t]$ and aligned prediction $\tilde{P}_t = [\tilde{R}_t \mid \tilde{\mathbf{t}}_t]$, we define

$$d_R(R_t, \tilde{R}_t) = \frac{180}{\pi} \cos^{-1} \left[\text{clip} \left(\frac{\text{tr}(R_t^\top \tilde{R}_t) - 1}{2}, -1, 1 \right) \right], \quad (9)$$

$$\text{RotErr} = \frac{1}{T} \sum_{t=1}^T d_R(R_t, \tilde{R}_t), \quad (10)$$

$$\text{TransErr}_{\text{Sim}(3)} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{c}_t - \tilde{\mathbf{c}}_t\|_2, \quad (11)$$

$$\text{CamMC}_{\text{Sim}(3)} = \frac{1}{T} \sum_{t=1}^T \|P_t - \tilde{P}_t\|_F, \quad (12)$$

where $\mathbf{c}_t$ is the camera center. Lower values are better for all three metrics.

Metric-scale trajectory evaluation. Sim(3) removes global scale, so we additionally evaluate absolute translation. Depth Anything 3 [38] with the DA3NESTED-GIANT-LARGE-1.1 metric-depth checkpoint estimates

depth for the first 20 stride-4 frames, covering 4.75 seconds. It receives RGB frames only and does not access target poses. After resizing the VGGT-Ω and DA3 depth maps to a common resolution, we fit a zero-intercept scale over valid pixels $\mathcal{V}$:

$$s_{\text{DA3}} = \underset{s}{\text{argmin}} \sum_{p \in \mathcal{V}} \|s d_{\text{VGGT}}(p) - d_{\text{DA3}}(p)\|_2^2. \quad (13)$$

Valid pixels have finite positive depth, DA3 depth below 200 m, and VGGT-Ω confidence above the frame median. We apply this scale once to the complete predicted trajectory and then permit SE(3) alignment only. We report

$$\text{ATE-RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\mathbf{c}_t - \hat{\mathbf{c}}_t\|_2^2}, \quad (14)$$

$$\text{TransErr}_{\text{Abs}} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{c}_t - \hat{\mathbf{c}}_t\|_2, \quad (15)$$

$$\text{ScaleRelErr} = \left| \frac{s_{\text{DA3}}}{s_{\text{oracle}}} - 1 \right| \times 100\%, \quad (16)$$

where s_{oracle} is used only to measure scale error and is never applied to the prediction.

Overall results. WorldCore gives the lowest error for every relative and metric-scale measure in Table 4. Its RotErr is 13.54°, compared with 19.88° for the second-best Matrix-Game 3.5, while its relative TransErr and CamMC are 1.475 and 1.546. After metric-scale calibration, WorldCore obtains 2.513 m ATE-RMSE and 2.290 m TransErr, compared with 3.819 m and 3.412 m for the second-best HY-WorldPlay. Absolute scale remains challenging for all methods, but WorldCore records the lowest ScaleRelErr at 129.47%.

Results by trajectory. Table 5 shows that WorldCore has the lowest RotErr on Trajectories 01, 02, 03, and 05, and ranks second on Trajectory 04. On the 103-second memory trajectory, its 9.06° error is substantially below the second-best 21.04°. The revisit gains in Table 3 therefore do not arise from suppressing camera motion or failing to complete the closed loop.

4.4. Visual Quality

Protocol. Following SANA-WM [87], we run VBench [24] in custom-input mode on the complete normalized videos. We report Subject Consistency (SC), Background Consistency (BC), Temporal Flickering (TF), Motion Smoothness (MS), Aesthetic Quality (AQ), Imaging Quality (IQ), Dynamic Degree (DD), and Overall Consistency (OC). SC, BC, AQ, and IQ use all retained frames; TF and MS use all valid adjacent pairs and triplets. DD evaluates optical flow on an 8-fps grid, while OC uses the official ViCLIP weights and eight-frame middle sampling against the scene

Table 4. Camera-control accuracy over 725 videos. Relative metrics use Sim(3) alignment; absolute metrics use metric-depth scale calibration followed by SE(3) alignment. Lower is better for every metric.

Method	Relative trajectory			Metric-scale trajectory		
	RotErr (°)	TransErr	CamMC	ATE-RMSE (m)	TransErr (m)	ScaleRelErr (%)
WorldCore	13.5357	1.4746	1.5462	2.5133	2.2903	129.47
DreamX-World	54.1163	2.7589	3.1380	7.4182	6.5104	670.20
EVOKE	26.0420	2.0417	2.1987	6.5000	5.8464	642.27
HY-WorldPlay	34.0506	2.1460	2.3585	<u>3.8192</u>	<u>3.4121</u>	<u>204.86</u>
JoyAI-Echo	21.4551	2.0723	2.1885	5.1619	4.6803	414.89
LingBot-World-v2	30.6145	2.0042	2.1922	13.2251	11.9363	944.78
Matrix-Game 3.5	<u>19.8808</u>	1.9201	2.0278	7.5411	6.6620	684.42
SANA-WM	23.5305	<u>1.7403</u>	<u>1.8866</u>	7.6170	6.9232	596.45

Table 5. RotErr in degrees for each trajectory. Values average 145 scenes; lower is better.

Method	01	02	03	04	05
WorldCore	8.0692	17.8531	9.6004	<u>23.0986</u>	9.0570
DreamX-World	29.3248	75.2496	70.8536	55.8628	39.2907
EVOKE	15.5088	29.2057	35.6112	24.6110	25.2734
HY-WorldPlay	22.5736	49.2396	25.6310	51.7706	<u>21.0379</u>
JoyAI-Echo	12.7201	25.7312	24.9603	18.7853	25.0785
LingBot-World-v2	22.8540	31.9535	18.2263	34.5799	45.4586
Matrix-Game 3.5	14.8556	<u>20.5969</u>	<u>13.0956</u>	26.9622	23.8937
SANA-WM	<u>9.5027</u>	22.6630	20.7919	36.3470	28.3479

description. VBench Overall aggregates the seven quality dimensions from SC through DD after normalization to the official bounds:

$$\text{Overall} = 100 \frac{\sum_k w_k \text{clip}((x_k - l_k)/(u_k - l_k), 0, 1)}{\sum_k w_k}, \quad (17)$$

where DD has weight 0.5 and the other dimensions have unit weight. OC is reported separately and does not enter Overall.

Results. WorldCore reaches 81.02 Overall in Table 6, ranking second and trailing EVOKE by 0.57 points. It ranks first on TF, MS, and OC, and second on SC and BC. Its AQ and IQ are 59.52 and 67.86, below the strongest frame-level scores, while DD remains high at 96.69. Thus, the memory and camera-control gains do not come at the cost of temporal quality, and the strong consistency scores are not explained by static outputs.

Table 6. VBench on 725 complete videos. Scores are percentages and higher is better. Overall aggregates the seven quality dimensions from SC through DD; OC is reported separately.

Method	SC	BC	TF	MS	AQ	IQ	DD	OC	Overall
WorldCore	81.80	90.07	95.94	98.77	59.52	67.86	96.69	26.69	81.02
DreamX-World	76.94	88.10	95.36	98.12	57.04	60.54	99.86	25.16	77.89
EVOKE	81.26	89.37	95.82	98.50	61.18	71.59	98.90	25.57	81.59
HY-WorldPlay	74.97	86.55	95.30	97.53	52.47	61.18	97.66	20.99	76.10
JoyAI-Echo	81.82	90.30	95.74	96.63	61.05	66.48	99.03	25.99	80.07
LingBot-World-v2	75.83	87.90	93.91	96.60	60.09	68.66	100.00	25.92	77.97
Matrix-Game 3.5	70.86	84.08	94.46	98.09	57.83	69.28	100.00	19.53	77.05
SANA-WM	79.73	89.87	95.69	98.65	62.73	66.51	99.17	25.43	80.92

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